

SV-16 microcontroller — build plan for a 4-person team

What this is: one path from where the project is today (RTL done, simulated, bitstream builds, no hardware) to a working hand-held microcontroller board — split into roles, work packages and gates so four people can work in parallel without blocking each other.

What it is not: a schedule promise. The durations assume 4 people at roughly **8-10 hours a week each** (evenings + one weekend half-day) and no prior PCB experience. Compress or stretch \$9 to your reality; the order and the gates matter more than the dates.

Current position (measured, not estimated):

SoC / CPU / peripherals	26 RTL files, 18 test suites, 473 checks, 0 failures (make test)
FPGA fit	9,191 LUT4 = 75.7 % of the LFE5U-12F, 18/32 block RAM, 45.45 MHz against a 37.5 MHz target
Firmware	ROM monitor in flash-resident boot ROM, A/B image slots with rollback, UART update at ~5 KB/s
Hardware	none. The board is specified in BOARD.md (34 pages, India-sourced BOM) — nothing ordered or built
Biggest unknowns	first-board bring-up, the 0.5 mm TQFP-144 assembly, and FPGA availability in India

0. Finish line — pick one before you start

Level	Definition	Reached at
L1 Demo	the real monitor runs in a terminal against the whole SoC, you can upload firmware and boot it — no hardware at all	end of Phase 1 (week 3)
L2 Prototype	a physical board: powered, configured, talks on USB, accepts an A/B firmware update over the wire, blinks the LEDs, drives a motor, survives a 24 h soak	end of Phase 4 (week 12)
L3 Product	the gap list in docs/MCU_READINESS.md closed: signed updates, fault record, debug probe, measured power, EMC, enclosure	Phase 5 (open-ended)

Tell your team which level you are aiming at. Everything below targets **L2**, with L3 items parked in Phase 5 so nobody starts them early.

1. The four roles

One person owns each area — a single writer per directory. Reviewing each other's work is mandatory; editing each other's files is not.

Role	Owens (paths)	Must be good at / learn	Shows up in
P1 — Systems / RTL lead ("the chip")	rtl/, constraints/, Makefile, the bitstream flow, integration	SystemVerilog, Yosys/nextpnr, ECP5 primitives + PLL, timing, the memory map	every phase; integrator for merges

Role	Owns (paths)	Must be good at / learn	Shows up in
P2 — Firmware / toolchain ("the code")	<code>firmware/</code> , <code>scripts/</code> (assembler, monitor host tool, packer), the C toolchain	Python, SV-16 ISA, UART protocols, later a C runtime; comfortable reading RTL	every phase
P3 — Hardware / PCB ("the board")	<code>BOARD.md</code> , <code>board/</code> , a new <code>hardware/</code> directory with the KiCad project, the BOM, bench work	Schematic capture + layout (KiCad), ECP5 configuration rules, hand soldering/rework, DMM/scope	weeks 1-10, heaviest in 1-4
P4 — Verification / release ("the referee")	<code>simulation/</code> , <code>docs/</code> , the acceptance matrix, tags/releases, CI-ish scripts	Verilator testbenches, writing acceptance criteria that can fail, documentation discipline, scepticism	every phase; gatekeeper for "done"

Why this split: it matches the four real bottlenecks (silicon, software, board, evidence) and each role has an independent deliverable that can be shown on a Friday. Two people never wait on one file.

If someone is missing for a week: P1 and P2 can cover each other; P4 is the one role that must not be skipped, because "the design is done" is only true if someone independent reproduced it (§6).

2. The path in five phases

Phase	Weeks	Goal	Gate (must pass to leave the phase)
0 — Ground truth	0 (2-3 days)	everyone can build, test and read the project	4 laptops each run <code>make test</code> → 18 suites, 0 failures; roles agreed; parts ordered
1 — Software traction + board definition	1-3	the machine becomes touchable; the schematic exists	G1: <code>make vboard</code> demo — type <code>?</code> , upload a firmware image, verify, boot it, all in a terminal. Schematic ERC-clean and reviewed
2 — Compiler + layout	3-6	you can write C for it; the PCB exists in KiCad	G2: gerbers released after review; C conformance suite green; V1 pin decision merged
3 — Fab + debug	6-9	boards on the bench; debugging works	G3: power-up checks 1-5 of <code>BOARD.md</code> §10 pass on the first assembled board
4 — Bring-up	9-12	the whole acceptance matrix passes	G4: <code>BOARD.md</code> §10 steps 6-14 signed off, v1.0 tagged, lessons written
5 — Product hardening	after	close the readiness gaps	each L3 item gets its own spec + test, like everything else here

Phase 0 — Ground truth (days, not weeks)

Who	Task	Done when
all	clone, source <code>scripts/</code> <code>sv16_venv.sh</code> , <code>make test</code> , <code>make lint</code> , <code>make bitstream</code>	each person has seen the numbers in <code>REPORT.md</code> come out of their own machine
all	read, in this order: <code>README.md</code> , <code>docs/MCU_READINESS.md</code> , <code>BOARD.md</code> §1-§4, <code>REPORT.md</code> §8	a 30-minute meeting where each person explains one section to the others

Who	Task	Done when
P3	order: 2 × LFE5U-12F-6TG144C (or speed 7), an FT232H/FT2232H JTAG adapter, a ₹100 QFP-144 practice breakout, solder paste + flux + wick	tracked order numbers in the repo (hardware/ORDER.md)
P4	set up the shared evidence folder: every Friday's log, screenshot or waveform goes in one place	a link the team can open

The FPGA order is the **longest-lead item on the whole project** (40+ weeks when the distributors are empty). Place it in week 1 and treat it as the project's pacemaker.

Phase 1 (weeks 1-3) — software traction, board definition

Person	Work packages	Hand-over at the end of the phase
P1	WP2 virtual board (S0), WP8 V1 pin decision	make vboard runs the real monitor in a terminal; board/TQFP144_PINOUT.md regenerated for the final pin map
P2	WP3 upload throughput (S1), WP4 monitor protocol freeze	≥20 KB/s upload path + sv16_mon.py --fast, with fallback to today's c command
P3	WP9 schematic capture, WP10 BOM quotes + FPGA order	hardware/sv16_board/*.kicad_sch, ERC clean, reviewed by P1
P4	WP5 host-tool unit tests, WP14 acceptance matrix draft	tests/ with golden assembler corpus + packer round-trip; docs/ACCEPTANCE.md first cut

Gate G1 demo script (run it in front of each other, not "it works on my machine"):

```
make firmware && make vboard           # terminal: monitor banner, '?' lists commands
# in the vboard terminal: upload the motor app, verify, boot, watch it run
make test                               # 18 suites, 0 failures
```

Phase 2 (weeks 3-6) — compiler and layout

Person	Work packages	Output
P1	WP8b V1 remap implementation + rebuilt bitstream, WP6 UART GDB stub (S3a)	new LPF, regenerated pin docs, gdb can attach and set a breakpoint
P2	WP5b C toolchain S2a-S2d (scripts/sv16_cc.py, crt0, printf, make cc)	make cc C=firmware/apps/motor.c produces an image + map
P3	WP11 layout, WP12 DFM review, order boards	4-layer, DRC clean, review checklist signed by P1 and P4
P4	WP5c C conformance suite (c_tb), WP14b soak scripts	every operator/control-flow/pointer case executed in simulation with expected values

Layout cannot start before the pin map is frozen (§4) and the schematic is reviewed — that is the one hard serialisation in this project.

Phase 3 (weeks 6-9) — fab and debug

Person	Work packages	Output
P3	assembly of 2 of the 5 boards (P1 helps), bring-up steps 1-5	one board configured over JTAG, console printing

Person	Work packages	Output
P1	WP7 JTAG debug bridge (S3b, JTAGG ER1), WP15 capacity work (S5)	OpenOCD can halt/peek/poke the CPU
P2	WP5d runtime polish, docs for the C flow	firmware/README section a beginner can follow
P4	WP16 docs + release process, acceptance matrix ready to execute	a checklist that a non-author can run on the board

Phase 4 (weeks 9-12) — bring-up

Everyone on one board at a time; P3 drives the bench, P1 owns the bitstream, P2 owns the firmware, P4 owns the evidence. Work the `BOARD.md` §10 list in order: rails → JTAG detect → configure → console → reset cause → LEDs → application flash → A/B update → rollback → motor → current → 24 h soak.

Expect this phase to find 3–6 real defects; that is the phase working correctly, not failing.

3. Work packages (the assignment table)

Effort is in **person-days**; "blocked by" is what must exist first.

ID	Work package	Owner	Effort	Blocked by	Definition of done
WP1	Toolchain on 4 machines, baseline green	all	0.5	—	everyone runs <code>make test (18/473/0)</code> and <code>make bitstream</code>
WP2	Virtual board (S0) : Verilator SoC + flash model + UART bridge to a terminal, <code>make vboard, sv16_mon.py --transport sim</code>	P1, P2	5	WP1	<code>make vboard</code> demo of §2 G1, plus one scripted session in <code>simulation/regression/</code>
WP3	Upload throughput (S1) : binary <code>u</code> command, block acks, optional 460800 baud, CRC per block	P2	4	WP1	≥20 KB/s measured and printed by the host tool; old <code>c</code> path still works; numbers into <code>docs/VERIFICATION.md</code>
WP4	Monitor protocol frozen: one page in <code>docs/BOOT_AND_PROGRAMMING.md</code> with every command, field and error	P2	1	WP3	P4 can implement a test client from the page alone
WP5	Host-tool unit tests : assembler golden corpus, packer round-trip, monitor framing, hex parser limits	P4	3	WP1	<code>python3 -m unittest</code> in <code>CI-ish</code> script, run by <code>make test</code>
WP6	C toolchain (S2) : ABI ADR, <code>scripts/sv16_cc.py</code> , <code>crt0</code> , <code>printf</code> , <code>memcpy</code> , <code>make cc</code>	P2	12	WP4	a real C program (motor + PWM + timer + IRQ) compiles, uploads and runs

ID	Work package	Owner	Effort	Blocked by	Definition of done
WP7	C conformance suite: every operator, control-flow shape, pointer case, call depth	P4	5	WP6	c_tb in make test, expected values checked in simulation
WP8	Pin map freeze + V1 remap: move gpio_a[1]/[2]/[4]/[6] off the Master-SPI pads, update LPF + docs + bitstream	P1	2	—	scripts/ sv16_board_pins.py --check passes; new bitstream with ≥ 44 MHz Fmax; BOARD.md §6.4 rewritten
WP9	Schematic capture from BOARD.md §4–§7	P3	6	WP8	ERC clean, netlist exported, P1 + P4 review checklist signed
WP10	BOM, quotes, orders (FPGA, adapter, practice kit, tools)	P3	2	—	order numbers in hardware/ORDER.md, total $\leq$ budget (\$8)
WP11	PCB layout (4-layer, power + FPGA + USB)	P3	8	WP9	DRC/DFM clean, gerbers + BOM + pick-and-place exported; thermal + decoupling rules of BOARD.md §11 applied
WP12	Fab + assembly plan: 5 boards, stencil, paste, hot-air, inspection	P3, P1	3	WP11	2 boards fully assembled and inspected, 3 bare as spares
WP13	Bring-up kit: bench checklist, current limits, test points, order of fitting parts	P3, P4	2	WP9	BOARD.md §10 steps 1–5 executable by someone who did not write it
WP14	Debug: S3a UART GDB stub, then S3b JTAGG bridge for OpenOCD	P1, P2	8	WP2	halt/step/peek/poke over UART; JTAG bridge demonstrated on hardware
WP15	Capacity (\$5): RAM remap, more SRAM, ROM vector table	P1	5	WP2	memory map updated, loader still boots both slots, tests green
WP16	Docs + release: ACCEPTANCE.md, beginner programming guide, v1.0 tag, lessons	P4	4	Phase 4	a stranger can rebuild the bitstream, flash the board and update firmware from the docs alone

Sum: ≈ 71 person-days ≈ 18 person-weeks — about 4.5 weeks of full-time work for four people, or ~ 11 calendar weeks at 8–10 h/week each, which is why §2 lands on week 12. The hardware chain (WP9→WP11→WP12→bring-up) is ~ 25 of those days and runs mostly on P3.

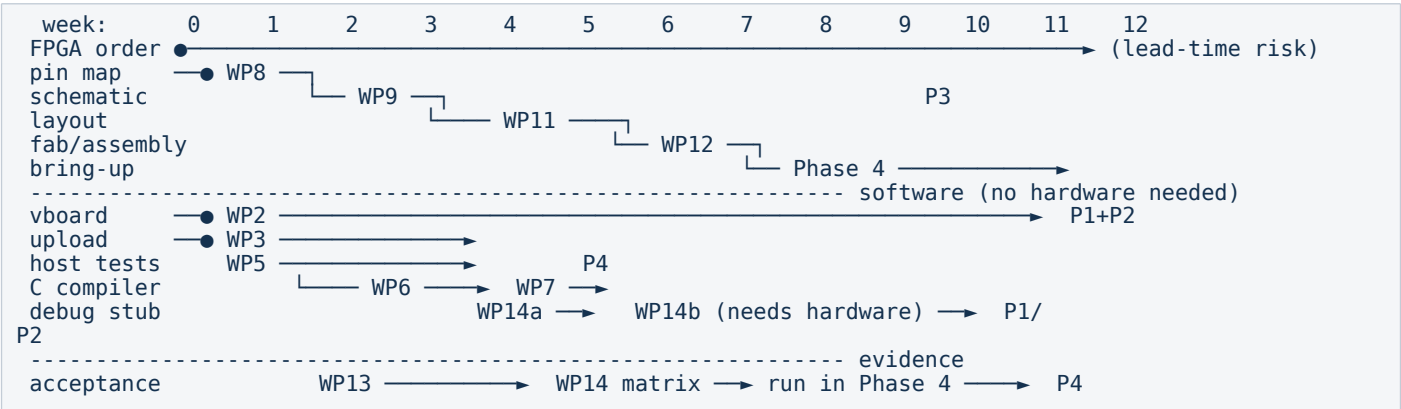
4. Interfaces that must freeze (the reason 4 people can work in parallel)

Write these down once and treat a change as a project event, not a commit:

Contract	File of record	Owner	Freeze by	If it changes
Pin map (signal → TQFP pin)	constraints/ ecp5_144tqfp.lpf + generated board/TQFP144_PINOUT.md	P1	end of week 2	P3 stops layout; regenerate docs; re-run timing
Memory map / MMIO	docs/MEMORY_MAP.md	P1	already frozen	needs an ADR + firmware change
Image + slot format	docs/ BOOT_AND_PROGRAMMING.md	P2	end of week 2	host tool, bootloader and images must move together
Monitor protocol	docs/ BOOT_AND_PROGRAMMING.md	P2	with WP4	version the change; keep the old command working
Connectors & pinouts	BOARD.md §7.5, §8	P3	before layout	reprint the schematic page + silkscreen
Power rails & sequencing	BOARD.md §4	P3	before layout	re-verify bring-up step 3
Acceptance criteria	docs/ACCEPTANCE.md	P4	before boards arrive	re-run the matrix

Rule of thumb: **anything a second person depends on gets a file of record and an owner.** If it is not written down, it is not frozen, and the other three are guessing.

5. Critical path and what can run in parallel



Serial (do not parallelise): pin map → schematic → layout → fab → bring-up. **Fully parallel:** vboard, upload throughput, host tests, C compiler, debug stub, docs. **Never** start layout before the pin map is frozen, and never order boards before the review checklist is signed — a respin costs a fortnight and ₹4,000.

6. Working agreements (this is what keeps 4 people unblocked)

- One writer per directory.** Reviewers comment in the PR; they do not push into someone else's WP.
- make test is the merge gate.** 18 suites, 0 failures, plus `make lint` (26 RTL files + pin-map drift check). A red build is fixed before anything new starts.

3. **No unverified numbers.** Every figure in a doc comes from a log, a datasheet or a bench measurement — this repo's existing culture (and it has already caught three invented ones).
4. **Short branches off main:** wp08-pin-remap, wp11-layout... merged by PR with one reviewer from another workstream. Delete the branch after merge.
5. **Friday demo, 20 minutes.** Each person shows one artifact: a terminal session, a schematic page, a test log, a scope screenshot. "I worked on it" is not a demo.
6. **Monday 30-minute replan.** Re-estimate the three WPs in flight; kill anything not on the critical path; move one WP if someone is blocked for more than a day.
7. **WIP limit of one.** Nobody carries two work packages at once. The board design is one WP, not five.
8. **Blame the process, then the part.** When bring-up fails, first check the schematic and the checklist, then the component — the odds are with the process.

7. What to buy in week 1 (in ₹, estimates)

Item	Why	₹
2 × FPGA (LFE5U-12F-6TG144C, or speed 7 if cheaper/in stock)	one to build, one spare — a dead FPGA otherwise costs another 40-week lead time	3,300
FT232H / FT2232H JTAG adapter	configure + debug over JTAG; also your first real "does the toolchain see the chip" test	700–1,500
Practice kit: QFP-144 breakout + spare 0.5 mm parts + flux + wick	learn drag-soldering before the ₹1,650 chip is involved	400
Solder paste + hot-air station (if not owned)	the only sane way to solder a TQFP-144 at home	2,500
Multimeter (if not owned) + USB logic analyser (₹700)	rails and UART are the two things you will debug most	1,500
Bench supply with current limit (if not owned)	bring-up without a current limit is how boards die	2,000
5 PCBs (4-layer, ~100 × 100 mm) + stencil, from JLCPCB	2 to assemble, 3 spares	3,500
Components for 5 boards (India-sourced, BOARD.md §3)	see the module-first option in BOARD.md §3.5	5,000
0603 passives assortment + wire/flux/headers/consumables	you will lose, short and re-order parts	1,200

≈ ₹21,000 with tools from scratch, ≈ ₹12,000 if your lab already has a meter, iron and air.

Everything except the FPGA can be ordered from Robu/Zbotic/Quartz; the FPGA is digikey.in / mouser.in / in.element14.com (BOARD.md §3.4).

8. Risks and who watches them

#	Risk	Own	Trigger to act	Mitigation
R1	FPGA not in stock (40+ week lead)	P3	no stock at digikey.in this week	accept speed grade 7 or 1 grade (verified: builds and closes at 49.53 MHz), or buy a Colorlight 5A-75B (₹1,500–3,000) to keep the RTL work on real silicon while waiting
R2	QFP-144 assembly fails	P3	first practice breakout has bridges you cannot clear	practice first, stencil + paste + hot air, inspect at 10×, assemble only 2 boards; fall back to BOARD.md §3.5 modules
R3	First power-up kills the FPGA	P3	1.1 V rail outside 1.045–1.155 V	bring-up step 2: fit rails, measure the 1.1 V divider, then fit U1; always use a current-limited supply
R4	Power-up order wrong on the bench	P3/P1	scope shows 1V1 before 3V3	use the PROGRAMN-hold fallback (BOARD.md §4.3); a host PWM/DTR pulse is enough
R5	Layout respin	P3/P4	review checklist incomplete	5 boards ordered, review by two people, DRC + DFM, module-first fallback for the power section
R6	Team drifts / roles blur	all	two Fridays with no demo	WIP limit, Monday replan, 20-minute demos, one writer per directory
R7	Toolchain differences between laptops	P1	"works on mine"	scripts/sv16_venv.sh pins Verilator/Yosys/nextpnr/ecppack; everyone runs make test before a PR
R8	C compiler bug ships as a "hardware" bug	P4	board misbehaves with C firmware only	WP7 conformance suite is the gate; reproduce in simulation before touching the bench

9. Compressing or stretching the plan

Situation	What to do
4 people, ~4 h/week each	keep Phases 0–2 as-is (software + schematic), then do one board per 6 weeks. Gate G1 and G2 are the value; L2 slips to ~6 months
Only 2 people	P1 takes P4's verification for RTL, P2 takes it for firmware; drop WP15 (capacity) and WP14b (JTAG bridge), and start hardware after G1
6 people	split P3's role into schematic (one person) and layout (another — layout is 8 days on its own), and give the C toolchain a second person for the runtime/printf half

Situation	What to do
No PCB ambitions	stop after G1: you still have a fully programmable MCU you can type at and flash, running on a Colorlight board or a simulator — make vboard is the deliverable
Deadline is a college review/demo	demo G1 (make vboard + upload + boot) and show BOARD.pdf + the schematic — that is a complete story without waiting for silicon

10. What to read first (per role)

Role	Read	Then
P1	docs/MEMORY_MAP.md, docs/BUS_ARCHITECTURE.md, docs/RESET_AND_CLOCK.md, ADR-021/024 (PLL)	REPORT.md §8.2 (S0, S3, S5), simulation/ to see how a testbench is written here
P2	docs/ISA.md, docs/BOOT_AND_PROGRAMMING.md, firmware/monitor/monitor.s	scripts/sv16_mon.py, then the S1/S2 items in REPORT.md §8.2
P3	BOARD.md (all of it), board/TQFP144_PINOUT.md, docs/SYNTHESIS_AND_DEPLOYMENT.md (pin map)	ECP5 sysCONFIG (TN1260) and the KiCad ECP5 design rules; BOARD.md §11 before drawing anything
P4	docs/VERIFICATION.md, every testbench in simulation/, docs/MCU_READINESS.md	write the acceptance matrix from BOARD.md §10 before the boards arrive

11. Tell me these five things and I will re-cut the plan

- Hours per week per person, and the calendar deadline (if any).
- Who has what: soldering iron/hot air, DMM, scope, bench supply, FPGA/Arduino experience, KiCad or any CAD experience, C/Python level.
- Budget ceiling for parts + tools (§7 is ₹12–21k with/without tools).
- Do you have a college lab or a makerspace with a reflow oven, microscope or assembly service?
- Do you want the first board to be the **module-first** version (BOARD.md §3.5, safer, bigger) or the all-soldered version (smaller, riskier)?

With those answers this document becomes a dated sprint plan with names next to each WP — and I can cut it down to a one-page chart for your group.

Revision 1.0 — written against commit f57f32a (project state: RTL complete, 473 checks green, bitstream builds, no hardware). Effort estimates are for a first-time hardware team; the two numbers taken from real measurements are the 473 checks and the 49.53 MHz speed-7 timing run.